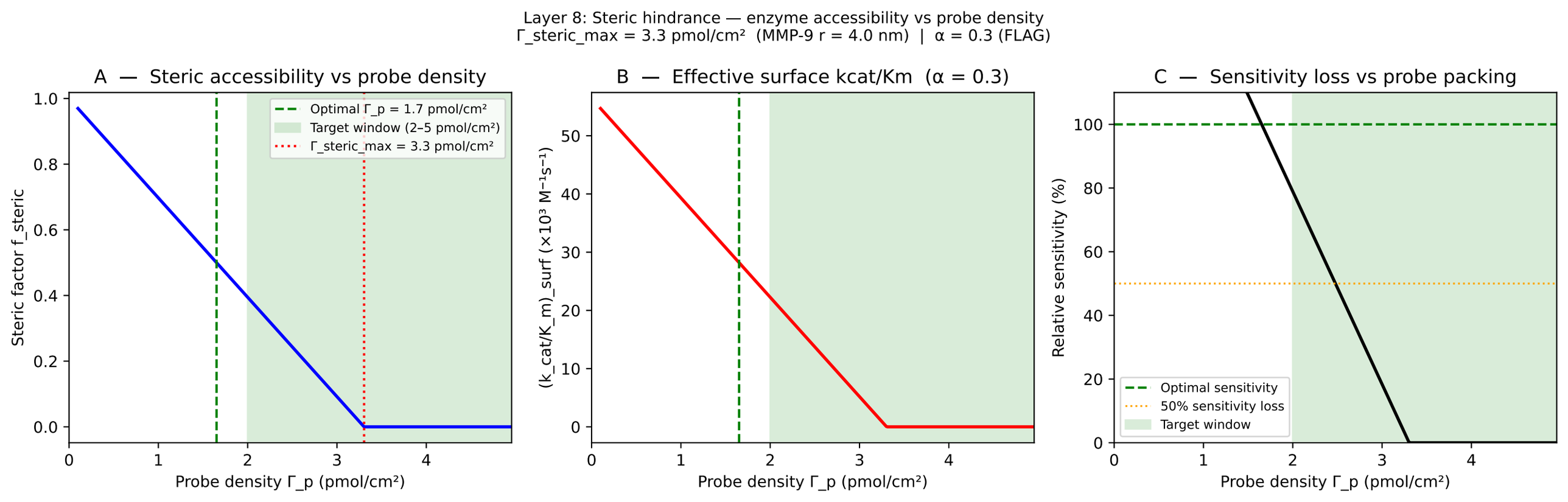
Layer 8: Steric hindrance — enzyme accessibility vs probe density
Γ_steric_max = 3.3 pmol/cm² (MMP-9 r = 4.0 nm) | α = 0.3 (FLAG)
A — Steric accessibility vs probe density
Optimal Γ_p = 1.7 pmol/cm²
Target window (2–5 pmol/cm²)
Γ_steric_max = 3.3 pmol/cm²
Steric factor f_steric
Probe density Γ_p (pmol/cm²)
B — Effective surface kcat/Km (α = 0.3)
(k_cat/K_m)_surf (×10³ M⁻¹s⁻¹)
Probe density Γ_p (pmol/cm²)
C — Sensitivity loss vs probe packing
Relative sensitivity (%)
Optimal sensitivity
50% sensitivity loss
Target window
Probe density Γ_p (pmol/cm²)